Linear Algebra and Its Applications

Section 1.2: Row Reduction and Echelon Forms

Detailed worked solutions

Mathjock

Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

Contents

1.2 Row reduction and echelon forms 3

Practice problem 1.2.P1: Describe every solution with one parameter 3

Practice problem 1.2.P2: Elimination with two free variables 3

Exercise 1.2.1: Distinguish echelon form from reduced form 4

Exercise 1.2.2: Check all echelon requirements 5

Exercise 1.2.3: Locate two pivot positions 5

Exercise 1.2.4: Locate three pivot positions 6

Exercise 1.2.5: All nonzero two-by-two echelon patterns 7

Exercise 1.2.6: All nonzero three-by-two echelon patterns 7

Exercise 1.2.7: Two pivots and one free variable 7

Exercise 1.2.8: A variable absent from both equations 8

Exercise 1.2.9: Interchange before eliminating 9

Exercise 1.2.10: A nonzero second pivot after elimination . . . 9
Exercise 1.2.11: Repeated equations leave two free variables . . . 10
Exercise 1.2.12: An augmented-column pivot rules out solutions . . . 11
Exercise 1.2.13: Five variables and two free choices . . . 11
Exercise 1.2.14: A zero basic variable simplifies the parameters . . . 12
Exercise 1.2.15: Read two symbolic echelon patterns . . . 13
Exercise 1.2.16: Contrast variable pivots with a constant pivot . . . 13
Exercise 1.2.17: A parameter in the constant column . . . 13
Exercise 1.2.18: A parameter that can remove one equation . . . 14
Exercise 1.2.19: Choose parameters for all three outcomes . . . 14
Exercise 1.2.20: A second three-outcome parameter family . . . 14
Exercise 1.2.21: Definitions of basic variables and solutions . . . 14
Exercise 1.2.22: Uniqueness of RREF and consistency cautions . . . 14
Exercise 1.2.23: Four coefficient pivots force uniqueness . . . 15
Exercise 1.2.24: No pivot in the last augmented column . . . 15
Exercise 1.2.25: A pivot in every coefficient row . . . 15
Exercise 1.2.26: Three pivots in a three-by-five coefficient matrix . . . 15
Exercise 1.2.27: Restate uniqueness in pivot language . . . 16
Exercise 1.2.28: Combine existence and uniqueness conditions . . . 16
Exercise 1.2.29: Why underdetermined systems cannot be unique . . . 16
Exercise 1.2.30: An inconsistent underdetermined example . . . 16
Exercise 1.2.31: A consistent overdetermined example . . . 16
Exercise 1.2.32: Estimate the backward share of reduction cost . . . 17
Exercise 1.2.33: Interpolate three points with a quadratic . . . 17
Exercise 1.2.34: Interpolate wind-tunnel measurements . . . 17

1.2 Row reduction and echelon forms

A pivot is a leading nonzero entry of an echelon row. In echelon form, zero rows are at the bottom, pivots move strictly to the right as one moves downward, and all entries below a pivot are zero. In reduced echelon form (RREF), every pivot is 1 and is the only nonzero entry of its column. RREF is unique even though the sequence of operations leading to it is not.

For a system, pivot variable columns correspond to basic variables, and nonpivot variable columns correspond to free variables. *First test consistency*: a pivot in the augmented column means no solution. Only a consistent system with a free variable has infinitely many solutions.

Practice problem 1.2.P1: Describe every solution with one parameter

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & -3 & -5 & 0 \\ 0 & 1 & 1 & 3 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-3)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & -2 & 9 \\ 0 & 1 & 1 & 3 \end{array}\right].$$

Read the equations, keeping every free variable.

The nonpivot variable is x_3. Set $x_3 = t_1$, where the parameters range independently over $\mathbb{R}$.

$$\begin{cases} x_1 = 2t_1 + 9 \\ x_2 = 3 - t_1 \\ x_3 = t_1 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. Infinitely many solutions, exactly as parametrized above.

Practice problem 1.2.P2: Elimination with two free variables

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 0 \\ -2 & 4 & 5 & -5 & 3 \\ 3 & -6 & -6 & 8 & 2 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-2)R_1$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 0 \\ 0 & 0 & 3 & 1 & 3 \\ 3 & -6 & -6 & 8 & 2 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (3)R_1$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 0 \\ 0 & 0 & 3 & 1 & 3 \\ 0 & 0 & -3 & -1 & 2 \end{array}\right].$$

Use $R_2 \leftarrow (\frac{1}{3})R_2$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 0 \\ 0 & 0 & 1 & \frac{1}{3} & 1 \\ 0 & 0 & -3 & -1 & 2 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-3)R_2$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 0 \\ 0 & 0 & 1 & \frac{1}{3} & 1 \\ 0 & 0 & 0 & 0 & 5 \end{array}\right].$$

Use $R_3 \leftarrow (\frac{1}{5})R_3$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 0 \\ 0 & 0 & 1 & \frac{1}{3} & 1 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (1)R_3$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 0 \\ 0 & 0 & 1 & \frac{1}{3} & 0 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-1)R_2$ to obtain

$$\left[\begin{array}{cccc|c} 1 & -2 & 0 & \frac{10}{3} & 0 \\ 0 & 0 & 1 & \frac{1}{3} & 0 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right].$$

Conclusion. A row with all variable coefficients zero and a nonzero constant asserts $0 = 1$ after scaling. This is impossible, so the system is inconsistent and its solution set is empty.

Exercise 1.2.1: Distinguish echelon form from reduced form

(a) Echelon only. The first pivot is 10, not 1, so reduced form fails, although pivots move right and their columns are zero below them.

(b) Reduced echelon form. The pivots are 1 in columns 1, 2, 4, and these pivot columns have no other nonzero entries. The 1 in column 3 of the first row does not matter because column 3 is not a pivot column.

(c) Neither. Row 3 is zero but row 4 is nonzero; zero rows must be last.

(d) Echelon only. The successive pivots $1, 2, 3, 4$ move right, but three pivots are not 1, and pivot columns may have entries above them.

Exercise 1.2.2: Check all echelon requirements

(a) Reduced echelon form: pivots in columns 1 and 2 are 1 and are the only nonzero entries in their columns; the zero row is last.

(b) Echelon only: the middle pivot is 2, so it is not reduced.

(c) Neither: the zero row is above nonzero rows.

(d) Echelon only: pivots in columns 2, 3, 5 move right and are 1, but there are nonzero entries above the latter pivots. A pivot being 1 by itself does not make a matrix reduced.

Exercise 1.2.3: Locate two pivot positions

What we must show. Reduce the matrix itself; it is not being declared an augmented matrix.

$$\begin{bmatrix} 1 & 2 & 4 & 8 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 9 & 12 \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - (2)R_1$ to obtain

$$\begin{bmatrix} 1 & 2 & 4 & 8 \\ 0 & 0 & -2 & -8 \\ 3 & 6 & 9 & 12 \end{bmatrix}.$$

Use $R_3 \leftarrow R_3 - (3)R_1$ to obtain

$$\begin{bmatrix} 1 & 2 & 4 & 8 \\ 0 & 0 & -2 & -8 \\ 0 & 0 & -3 & -12 \end{bmatrix}.$$

Use $R_2 \leftarrow (-\frac{1}{2})R_2$ to obtain

$$\begin{bmatrix} 1 & 2 & 4 & 8 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & -3 & -12 \end{bmatrix}.$$

Use $R_3 \leftarrow R_3 - (-3)R_2$ to obtain

$$\begin{bmatrix} 1 & 2 & 4 & 8 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Use $R_1 \leftarrow R_1 - (4)R_2$ to obtain

$$\begin{bmatrix} 1 & 2 & 0 & -8 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Conclusion. The pivot positions are $(1,1)$ and $(2,3)$ in both the final matrix and the original matrix; the pivot columns are columns 1 and 3.

The circled entries mark those positions explicitly:

$$\begin{bmatrix} \textcircled{1} & 2 & 4 & 8 \\ 2 & 4 & \textcircled{6} & 8 \\ 3 & 6 & 9 & 12 \end{bmatrix} \longrightarrow \begin{bmatrix} \textcircled{1} & 2 & 0 & -8 \\ 0 & 0 & \textcircled{1} & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

A pivot position refers to a location, not necessarily the same numerical entry before and after reduction.

Exercise 1.2.4: Locate three pivot positions

What we must show. Reduce the matrix itself; it is not being declared an augmented matrix.

$$\begin{bmatrix} 1 & 2 & 4 & 5 \\ 2 & 4 & 5 & 4 \\ 4 & 5 & 4 & 2 \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - (2)R_1$ to obtain

$$\begin{bmatrix} 1 & 2 & 4 & 5 \\ 0 & 0 & -3 & -6 \\ 4 & 5 & 4 & 2 \end{bmatrix}.$$

Use $R_3 \leftarrow R_3 - (4)R_1$ to obtain

$$\begin{bmatrix} 1 & 2 & 4 & 5 \\ 0 & 0 & -3 & -6 \\ 0 & -3 & -12 & -18 \end{bmatrix}.$$

Use $R_2 \leftrightarrow R_3$ to obtain

$$\begin{bmatrix} 1 & 2 & 4 & 5 \\ 0 & -3 & -12 & -18 \\ 0 & 0 & -3 & -6 \end{bmatrix}.$$

Use $R_2 \leftarrow (-\frac{1}{3})R_2$ to obtain

$$\begin{bmatrix} 1 & 2 & 4 & 5 \\ 0 & 1 & 4 & 6 \\ 0 & 0 & -3 & -6 \end{bmatrix}.$$

Use $R_3 \leftarrow (-\frac{1}{3})R_3$ to obtain

$$\begin{bmatrix} 1 & 2 & 4 & 5 \\ 0 & 1 & 4 & 6 \\ 0 & 0 & 1 & 2 \end{bmatrix}.$$

Use $R_1 \leftarrow R_1 - (4)R_3$ to obtain

$$\begin{bmatrix} 1 & 2 & 0 & -3 \\ 0 & 1 & 4 & 6 \\ 0 & 0 & 1 & 2 \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - (4)R_3$ to obtain

$$\begin{bmatrix} 1 & 2 & 0 & -3 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 2 \end{bmatrix}.$$

Use $R_1 \leftarrow R_1 - (2)R_2$ to obtain

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 2 \end{bmatrix}.$$

Conclusion. The pivot positions are $(1,1)$, $(2,2)$, and $(3,3)$ in both the final matrix and the original matrix; the pivot columns are columns 1, 2, and 3.

The original and reduced pivot positions can be circled as follows:

$$\begin{bmatrix} \textcircled{1} & 2 & 4 & 5 \\ 2 & \textcircled{4} & 5 & 4 \\ 4 & 5 & \textcircled{4} & 2 \end{bmatrix} \longrightarrow \begin{bmatrix} \textcircled{1} & 0 & 0 & 1 \\ 0 & \textcircled{1} & 0 & -2 \\ 0 & 0 & \textcircled{1} & 2 \end{bmatrix}.$$

Exercise 1.2.5: All nonzero two-by-two echelon patterns

Let ■ mean a nonzero pivot and $*$ an arbitrary entry. A nonzero 2×2 matrix has either one or two nonzero echelon rows. With one nonzero row, its pivot is in column 1 or column 2. With two, pivots must be in columns 1 then 2. Thus all possibilities are

$$\begin{bmatrix} ■ & * \\ 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & ■ \\ 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} ■ & * \\ 0 & ■ \end{bmatrix}.$$

There are no other choices: a pivot cannot occur to the left of the pivot above it, and zero rows cannot precede nonzero rows.

Exercise 1.2.6: All nonzero three-by-two echelon patterns

There can be at most two pivots because there are only two columns. As in Exercise 5, either the sole pivot is in column 1, the sole pivot is in column 2, or the two pivots occupy columns 1 and 2 in succession:

$$\begin{bmatrix} ■ & * \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & ■ \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} ■ & * \\ 0 & ■ \\ 0 & 0 \end{bmatrix}.$$

The third row must be zero in every case, and nonzeroness excludes the all-zero matrix. These exhaust the echelon possibilities.

Exercise 1.2.7: Two pivots and one free variable

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & 3 & 4 & 7 \\ 3 & 9 & 7 & 6 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (3)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 3 & 4 & 7 \\ 0 & 0 & -5 & -15 \end{array}\right].$$

Use $R_2 \leftarrow (-\frac{1}{5})R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 3 & 4 & 7 \\ 0 & 0 & 1 & 3 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (4)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 3 & 0 & -5 \\ 0 & 0 & 1 & 3 \end{array}\right].$$

Read the equations, keeping every free variable.

The nonpivot variable is x_2. Set $x_2 = t_1$, where the parameters range independently over $\mathbb{R}$.

$$\begin{cases} x_1 = -3t_1 - 5 \\ x_2 = t_1 \\ x_3 = 3 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. Infinitely many solutions, exactly as parametrized above.

Exercise 1.2.8: A variable absent from both equations

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & -3 & 0 & -5 \\ -3 & 7 & 0 & 9 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-3)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -3 & 0 & -5 \\ 0 & -2 & 0 & -6 \end{array}\right].$$

Use $R_2 \leftarrow (-\frac{1}{2})R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -3 & 0 & -5 \\ 0 & 1 & 0 & 3 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-3)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 3 \end{array}\right].$$

Read the equations, keeping every free variable.

The nonpivot variable is x_3. Set $x_3 = t_1$, where the parameters range independently over $\mathbb{R}$.

$$\begin{cases} x_1 = 4 \\ x_2 = 3 \\ x_3 = t_1 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. Infinitely many solutions, exactly as parametrized above.

Exercise 1.2.9: Interchange before eliminating

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 0 & 1 & -2 & 3 \\ 1 & -3 & 4 & -6 \end{array}\right].$$

Use $R_1 \leftrightarrow R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -3 & 4 & -6 \\ 0 & 1 & -2 & 3 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-3)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & -2 & 3 \\ 0 & 1 & -2 & 3 \end{array}\right].$$

Read the equations, keeping every free variable.

The nonpivot variable is x_3. Set $x_3 = t_1$, where the parameters range independently over $\mathbb{R}$.

$$\begin{cases} x_1 = 2t_1 + 3 \\ x_2 = 2t_1 + 3 \\ x_3 = t_1 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. Infinitely many solutions, exactly as parametrized above.

Exercise 1.2.10: A nonzero second pivot after elimination

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & -2 & -1 & 4 \\ -2 & 4 & -5 & 6 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-2)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -2 & -1 & 4 \\ 0 & 0 & -7 & 14 \end{array}\right].$$

Use $R_2 \leftarrow (-\frac{1}{7})R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -2 & -1 & 4 \\ 0 & 0 & 1 & -2 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-1)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -2 & 0 & 2 \\ 0 & 0 & 1 & -2 \end{array}\right].$$

Read the equations, keeping every free variable.

The nonpivot variable is x_2. Set $x_2 = t_1$, where the parameters range independently over $\mathbb{R}$.

$$\begin{cases} x_1 = 2t_1 + 2 \\ x_2 = t_1 \\ x_3 = -2 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. Infinitely many solutions, exactly as parametrized above.

Exercise 1.2.11: Repeated equations leave two free variables

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 3 & -2 & 4 & 0 \\ 9 & -6 & 12 & 0 \\ 6 & -4 & 8 & 0 \end{array}\right].$$

Use $R_1 \leftarrow (\frac{1}{3})R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -\frac{2}{3} & \frac{4}{3} & 0 \\ 9 & -6 & 12 & 0 \\ 6 & -4 & 8 & 0 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (9)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -\frac{2}{3} & \frac{4}{3} & 0 \\ 0 & 0 & 0 & 0 \\ 6 & -4 & 8 & 0 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (6)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -\frac{2}{3} & \frac{4}{3} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Read the equations, keeping every free variable.

The nonpivot variables are x_2, x_3. Set $x_2 = t_1$, $x_3 = t_2$, where the parameters range independently over $\mathbb{R}$.

$$\begin{cases} x_1 = \frac{2t_1}{3} - \frac{4t_2}{3} \\ x_2 = t_1 \\ x_3 = t_2 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. Infinitely many solutions, exactly as parametrized above.

Exercise 1.2.12: An augmented-column pivot rules out solutions

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{cccc|c} 1 & 0 & -9 & 0 & 4 \\ 0 & 1 & 3 & 0 & -1 \\ 0 & 0 & 0 & 1 & -7 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (4)R_4$ to obtain

$$\left[\begin{array}{cccc|c} 1 & 0 & -9 & 0 & 0 \\ 0 & 1 & 3 & 0 & -1 \\ 0 & 0 & 0 & 1 & -7 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-1)R_4$ to obtain

$$\left[\begin{array}{cccc|c} 1 & 0 & -9 & 0 & 0 \\ 0 & 1 & 3 & 0 & 0 \\ 0 & 0 & 0 & 1 & -7 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-7)R_4$ to obtain

$$\left[\begin{array}{cccc|c} 1 & 0 & -9 & 0 & 0 \\ 0 & 1 & 3 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right].$$

Conclusion. A row with all variable coefficients zero and a nonzero constant asserts $0 = 1$ after scaling. This is impossible, so the system is inconsistent and its solution set is empty.

Exercise 1.2.13: Five variables and two free choices

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccccc|c} 1 & -3 & 0 & -1 & 0 & -2 \\ 0 & 1 & 0 & 0 & -4 & 1 \\ 0 & 0 & 0 & 1 & 9 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-1)R_3$ to obtain

$$\left[\begin{array}{ccccc|c} 1 & -3 & 0 & 0 & 9 & 2 \\ 0 & 1 & 0 & 0 & -4 & 1 \\ 0 & 0 & 0 & 1 & 9 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-3)R_2$ to obtain

$$\left[\begin{array}{ccccc|c} 1 & 0 & 0 & 0 & -3 & 5 \\ 0 & 1 & 0 & 0 & -4 & 1 \\ 0 & 0 & 0 & 1 & 9 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

Read the equations, keeping every free variable.

The nonpivot variables are x_3, x_5. Set $x_3 = t_1$, $x_5 = t_2$, where the parameters range independently over $\mathbb{R}$.

$$\begin{cases} x_1 = 3t_2 + 5 \\ x_2 = 4t_2 + 1 \\ x_3 = t_1 \\ x_4 = 4 - 9t_2 \\ x_5 = t_2 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. Infinitely many solutions, exactly as parametrized above.

Exercise 1.2.14: A zero basic variable simplifies the parameters

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{cccccc|c} 1 & 0 & -5 & 0 & -8 & 3 \\ 0 & 1 & 4 & -1 & 0 & 6 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-8)R_3$ to obtain

$$\left[\begin{array}{cccccc|c} 1 & 0 & -5 & 0 & 0 & 3 \\ 0 & 1 & 4 & -1 & 0 & 6 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

Read the equations, keeping every free variable.

The nonpivot variables are x_3, x_4. Set $x_3 = t_1$, $x_4 = t_2$, where the parameters range independently over $\mathbb{R}$.

$$\begin{cases} x_1 = 5t_1 + 3 \\ x_2 = -4t_1 + t_2 + 6 \\ x_3 = t_1 \\ x_4 = t_2 \\ x_5 = 0 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. Infinitely many solutions, exactly as parametrized above.

Exercise 1.2.15: Read two symbolic echelon patterns

(a) Two pivots among three variable columns.

The last row is zero, and the augmented column is not a pivot column. Thus the system is consistent. Columns 1 and 2 are pivot columns; column 3 supplies one free variable. Every choice of that variable allows back substitution, so there are infinitely many solutions.

(b) Three pivots among four variable columns.

The pivot columns are 2, 3, 4; column 5 is the augmented column and has no pivot. Thus the system is consistent. The first variable is free, so the solution is not unique: there are infinitely many solutions.

Conclusion. Both systems are consistent and neither has a unique solution.

Exercise 1.2.16: Contrast variable pivots with a constant pivot

(a) A pivot in the constant column.

There are three columns total, so column 3 is augmented. The last row is $[0\ 0 \mid \blacksquare]$, where $\blacksquare \neq 0$. It asserts $0 = \blacksquare$, a contradiction. The system is inconsistent.

(b) No pivot in the constant column.

Column 5 is augmented and has no pivot; the pivots are in columns 1, 3, 4. The system is consistent, and variable 2 is free. It consequently has infinitely many solutions.

Conclusion. Part (a) has no solution; part (b) is consistent but not unique.

Exercise 1.2.17: A parameter in the constant column

Eliminate the first coefficient in row 2 by $R_2 \leftarrow R_2 + 2R_1$. The last row becomes $[0\ 1 \mid h + 8]$. Its variable coefficient is 1 for every h, so it never produces a contradiction. Set $x_2 = h + 8$ and then $x_1 = 4 + x_2 = h + 12$.

Conclusion. The system is consistent for every real h.

Exercise 1.2.18: A parameter that can remove one equation

Use $R_2 \leftarrow R_2 - hR_1$. Its result is $[0\ 6+3h \mid -2-h]$. If $h \neq -2$, the variable coefficient is nonzero and gives a solution by back substitution. If $h = -2$, both entries are zero, so the second row is merely $0 = 0$; solutions still exist.

Conclusion. All real h work. At $h = -2$ the solutions are not unique; at every other h there is a unique solution.

Exercise 1.2.19: Choose parameters for all three outcomes

Elimination gives

$$\begin{bmatrix} 1 & h & 2 \\ 0 & 8-4h & k-8 \end{bmatrix}.$$

If $h \neq 2$, the second coefficient is nonzero, giving a unique solution for every k. If $h = 2$, the last equation is $0 = k - 8$. It is impossible for $k \neq 8$ and imposes no constraint for $k = 8$, leaving x_2 free.

Conclusion. (a) No solution: $h = 2, k \neq 8$, for example $(2, 0)$. (b) Unique: $h \neq 2$, any k, for example $(0, 0)$. (c) Infinitely many: $h = 2, k = 8$.

Exercise 1.2.20: A second three-outcome parameter family

Replace row 2 by row 2 minus twice row 1. The last row is $[0\ h+6 \mid k-2]$. If $h \neq -6$ it determines x_2 uniquely. If $h = -6$, it is compatible precisely when $k = 2$, and then x_2 is unrestricted.

Conclusion. (a) No solution: $h = -6, k \neq 2$, for example $(-6, 0)$. (b) Unique: $h \neq -6$, any k, for example $(0, 0)$. (c) Infinitely many: $h = -6, k = 2$.

Exercise 1.2.21: Definitions of basic variables and solutions

(a) False. The uniqueness-of-RREF theorem says every matrix has exactly one reduced echelon form; different operation sequences give that same matrix.

(b) False. Row reduction applies to any rectangular matrix, whether or not its last column represents constants in a system.

(c) True. A basic variable corresponds by definition to a pivot column of the coefficient matrix.

(d) True. Solving a system means describing all its solutions. A parametric description does this while recording the free choices.

(e) False. In $[0\ 0\ 0\ 5 \mid 0]$ the nonzero entry is in a variable column: the equation is $5x_4 = 0$, which is satisfiable.

Exercise 1.2.22: Uniqueness of RREF and consistency cautions

(a) True. This is precisely the uniqueness theorem for RREF.

(b) False. A pivot in *every* augmented-matrix column includes a pivot in the final column, implying an impossible equation. For example [0 | 1] already exhibits the obstruction (and an identity augmented matrix gives examples with every column a pivot column).

(c) False. Pivot positions are determined from the unique RREF, not by the particular path of row operations.

(d) True. A general solution explicitly describes all solutions, including the parameters and their allowed values.

(e) False. Consistency is also required. The system $0x_1 + 0x_2 = 1$ has nonpivot variable columns but no solutions.

Exercise 1.2.23: Four coefficient pivots force uniqueness

There are four variable columns and all four are pivot columns. In echelon form, the four pivots occupy all four rows, so the final augmented column cannot contain an additional pivot. Hence there is no contradiction row and the system is consistent. Since all variables are basic, no free choice remains; back substitution determines exactly one value for each variable.

Conclusion. The system has a unique solution.

Exercise 1.2.24: No pivot in the last augmented column

The fifth column is the constant column. By hypothesis it is not a pivot column. Thus RREF has no row $[0\ 0\ 0\ 0 \mid d]$ with $d \neq 0$, which is exactly the consistency criterion.

Conclusion. Yes, the system is consistent. It cannot have a unique solution: four variable columns but only three rows leave at least one free variable.

Exercise 1.2.25: A pivot in every coefficient row

A pivot in every coefficient-matrix row uses all available rows. Appending a right side cannot change the coefficient entries or create an additional row in which all coefficients vanish but the constant is nonzero. More concretely, apply to the right side the same row operations that put the coefficient matrix in echelon form. Every resulting row has a variable pivot, so no resulting row can assert $0 = d \neq 0$.

Conclusion. The system is consistent for every possible right side.

Exercise 1.2.26: Three pivots in a three-by-five coefficient matrix

There are three rows and three pivot columns. Different pivots occupy different rows, so there is a pivot in every coefficient row. By the argument in Exercise 25, there can be no contradiction row for any right side. There are five variables but only three pivots, so two variables are free.

Conclusion. The system is consistent and has infinitely many solutions.

Exercise 1.2.27: Restate uniqueness in pivot language

The exact completion is: "every column of the coefficient matrix is a pivot column." A nonpivot variable column is equivalent to a free variable. In a consistent system a free variable gives infinitely many solutions; without any free variables, all variables are fixed.

Conclusion. For a consistent system, uniqueness is equivalent to a pivot in every variable column.

Exercise 1.2.28: Combine existence and uniqueness conditions

Two facts are needed. The last (augmented) column must *not* be a pivot column, ensuring consistency. Every preceding (coefficient) column must be a pivot column, ensuring that there are no free variables. Conversely those two facts give existence and uniqueness by Theorem 2.

Conclusion. Pivots in every coefficient column, and no pivot in the augmented column. Neither requirement alone is sufficient.

Exercise 1.2.29: Why underdetermined systems cannot be unique

If there are m equations and n unknowns with $m < n$, there can be at most m pivots. Thus at least $n - m > 0$ variable columns are nonpivot. If the system is consistent, these give free variables and infinitely many solutions. If it is inconsistent, it has no solutions. Neither case has exactly one solution.

Conclusion. No underdetermined linear system can have a unique solution.

Exercise 1.2.30: An inconsistent underdetermined example

For example,

$$x_1 + x_2 + x_3 = 0, \qquad x_1 + x_2 + x_3 = 1.$$

There are only two equations for three unknowns. Subtracting the first from the second gives $0 = 1$, so the system is inconsistent.

Conclusion. Underdetermined does not mean consistent.

Exercise 1.2.31: A consistent overdetermined example

Yes. Consider

$$x_1 = 1, \qquad x_2 = 2, \qquad x_1 + x_2 = 3.$$

The pair $(1, 2)$ satisfies every equation, so this three-equation, two-variable system is consistent. The third equation is redundant: it is the sum of the first two, rather than an incompatible additional constraint.

Conclusion. An overdetermined system can be consistent; equation counts alone do not decide consistency.

Exercise 1.2.32: Estimate the backward share of reduction cost

The numerical note estimates the forward cost by $F_n = 2n^3/3 + n^2/2 - 7n/6$ and bounds the backward cost by $B_n = n^2$. Using the leading forward term, the approximate backward fraction of *total* cost is

$$\frac{n^2}{(2/3)n^3 + n^2} = \frac{3}{2n+3}.$$

At $n = 20$ this is $3/43 \approx 0.06977$, about 7%; at $n = 200$ it is $3/403 \approx 0.007444$, about 0.74%. Using the full stated forward count instead gives $400/(5510 + 400) \approx 6.77\%$ and $40000/(5353100 + 40000) \approx 0.742\%$. The cost of the backward phase is stated as an upper bound, so these are cost-model estimates, not an exact count for every input matrix.

Conclusion. About 7% for $n = 20$ and 0.74% for $n = 200$.

Exercise 1.2.33: Interpolate three points with a quadratic

The coefficient vector is (a_0, a_1, a_2). Its augmented matrix is

$$\begin{bmatrix} 1 & 1 & 1 & 6 \\ 1 & 2 & 4 & 15 \\ 1 & 3 & 9 & 28 \end{bmatrix}.$$

Use $R_2 \leftarrow R_2 - R_1$ and $R_3 \leftarrow R_3 - R_1$ to get rows [0 1 3 | 9] and [0 2 8 | 22] below the first. Then $R_3 \leftarrow R_3 - 2R_2$ gives [0 0 2 | 4]. Thus $a_2 = 2$, $a_1 = 9 - 3(2) = 3$, and $a_0 = 6 - 3 - 2 = 1$.

$$p(t) = 1 + 3t + 2t^2.$$

Check: $p(1) = 6$, $p(2) = 15$, and $p(3) = 28$.

Conclusion. The unique interpolating quadratic is $1 + 3t + 2t^2$.

Exercise 1.2.34: Interpolate wind-tunnel measurements

What we must show. Interpolate all six measured data points and convert the requested velocity into the table's units.

Set up six equations for six coefficients.

For each listed pair (t, F), use $a_0 + a_1 t + a_2 t^2 + a_3 t^3 + a_4 t^4 + a_5 t^5 = F$. The coefficient matrix has rows $[1, t, t^2, t^3, t^4, t^5]$ for $t = 0, 2, 4, 6, 8, 10$; the right side is $[0, 29/10, 74/5, 198/5, 743/10, 119]^T$. Treating the printed decimals as the given exact data, elimination gives

$$\begin{aligned} a_0 &= 0, & a_1 &= 137/80, & a_2 &= -1147/960, \\ a_3 &= 127/192, & a_4 &= -269/3840, & a_5 &= 1/384. \end{aligned}$$

Thus

$$p(t) = \frac{137t}{80} - \frac{1147t^2}{960} + \frac{127t^3}{192} - \frac{269t^4}{3840} + \frac{t^5}{384}.$$

Substitution at the six inputs gives exactly the six measured outputs, so the computed coefficients satisfy the full system. They are unique: a difference of two degree-at-most-five interpolants would have six distinct zeros and hence would be the zero polynomial.

Use the measurement units.

The velocity 750 ft/sec corresponds to $t = 7.5$, not $t = 750$. The force is tabulated in hundreds of pounds. Consequently

$$p(7.5) = \frac{132789}{2048} \approx 64.83838, \qquad \text{force} \approx 6483.84 \text{ lb}.$$

Explain why lowering the degree fails.

The unique degree-at-most-five interpolant has nonzero t^5 coefficient $1/384$. Thus no polynomial of degree less than five can pass through all six data points. For a direct cubic check, fitting the first four points gives

$$q(t) = \frac{13t^3}{160} + \frac{51t^2}{80} - \frac{3t}{20}.$$

But $q(8) = 81.2 \neq 74.3$ and $q(10) = 143.5 \neq 119$. The six-equation cubic system is inconsistent, rather than merely less accurate.

Conclusion. The displayed quintic interpolates the data; its prediction at 750 ft/sec is approximately 6484 lb.